Ex. 9.6

Q25

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1+n}{n^2}$$

Idea: • $\dfrac{1+n}{n^2} \sim \dfrac{1}{n} \Rightarrow \displaystyle\sum_{n=1}^{\infty} \left| (-1)^{n+1} \frac{1+n}{n} \right|$ div.

(not abs. conv.)

• Use AST to show $\displaystyle\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1+n}{n^2}$ conv.

(and so it conv. conditionally)

AST: on $\sum (-1)^n b_n$

1. $b_n \geq 0$ for $n \geq \boxed{\quad}$
2. $\{b_n\}$ is eventually decreasing $\left[\begin{array}{l} b_n \geq b_{n+1} \\ \text{for all } n \geq \underline{N} \end{array} \right]$ some positive integer
3. $b_n \to 0$

(AST) ① Clearly $\frac{1+n}{n^2} \geq 0$ for all $n \geq 1$.

② $\left\{\frac{1+n}{n^2}\right\}$ is decreasing on $\underline{n \geq 1}$.

- $\{a_n\}$ is inc. Then $\{\frac{1}{a_n}\}$ is dec.

- Find derivative of $\frac{1+x}{x^2}$ (Tedious)
- Sum of two decreasing func. is decreasing.

Draft: $\frac{1+n}{n^2} = \underset{\text{dec}}{\frac{1}{n^2}} + \underset{\text{dec}}{\frac{1}{n}} \Rightarrow \frac{1+n}{n^2}$ dec.

Since $\{\frac{1}{n^2}\}$ and $\{\frac{1}{n}\}$ are dec. on $n \geq 1$,

$\left\{\frac{1+n}{n^2}\right\}$ is dec. on $n \geq 1$

$\because \frac{1+n}{n^2} = \frac{1}{n^2} + \frac{1}{n}$

- $a_n \geq a_{n+1}$ for $n \geq 1$ [If the expression is "simple"]